

System integration is important because it connects systems together and adds value to a business. This helps to make things more efficient and improves the way data moves around. For example, system integration can link together things like inventory, CRM, and accounting systems. This means that people do not have to do much work by hand. System integration is about making it work better by connecting all these different systems, such as inventory, CRM, and accounting.

Star Integration

The star integration is also known as spaghetti integration. It is a way to connect every system to every other system. This is done by making point-to-point connections between the systems. When you draw a picture of this, it looks like a star or a big mess of spaghetti.

The star integration method makes the systems work better together. As you add more systems, it gets very complicated and hard to take care of. This means it costs a lot of money to keep it running.

For example, you have a CRM system, an inventory system, and an accounting system. The star integration method would connect each of these systems to every system. So the CRM system would send sales data to both the inventory system and the accounting system. This happens without using a hub to help the systems talk to each other. The CRM system, the inventory system, and the accounting system all exchange data with each other.

Horizontal Integration

Horizontal integration uses something called "middleware," like an enterprise service bus, to make things simpler. This Enterprise Service Bus acts like an interface for all the subsystems; it makes connections between them easier.

The good thing about this is that it reduces the number of direct links between systems. This means it lowers costs and makes it easier to swap out subsystems if you need to. You can use tools called adapters to help the subsystems talk to each other. These adapters can help with messaging or web services.

For example, an enterprise service bus can connect systems like CRM, ERP, and HR systems together. The CRM system can share sales data with the systems through the Enterprise Service Bus. This way, all the systems can see the sales data without having to be connected to the CRM system. Horizontal integration makes it easy for systems like CRM, ERP, and HR to work together.

Vertical Integration

Vertical integration puts systems into groups based on what they do, like human resources or finance. This way, people who are experts in these areas can work on them. It is easier to work with a few companies, but it can get very expensive to add new things. For example, a hospital can use integration to organize its patient data systems. They can connect the lab systems to a system in the Intensive Care Unit and then to the clinical records, all of which are part of the medical group. Vertical integration is used in this hospital example to make the systems work better.

Common Data Format

This method is really helpful because it creates a language that all systems can understand. It does this by setting up a shared format for data that any system can use. This means that each system only has to convert its data one time to fit this format. This is an improvement over other ways of doing things, where many systems have to convert data back and forth between each other.

For example, hospitals can use this method to make sure that patient information like name, age, and condition is the same in the lab and the intensive care unit. It can also be used to match zip codes with city names when different apps are talking to each other. The shared data format makes it easier for systems to work together and avoids a lot of work.

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